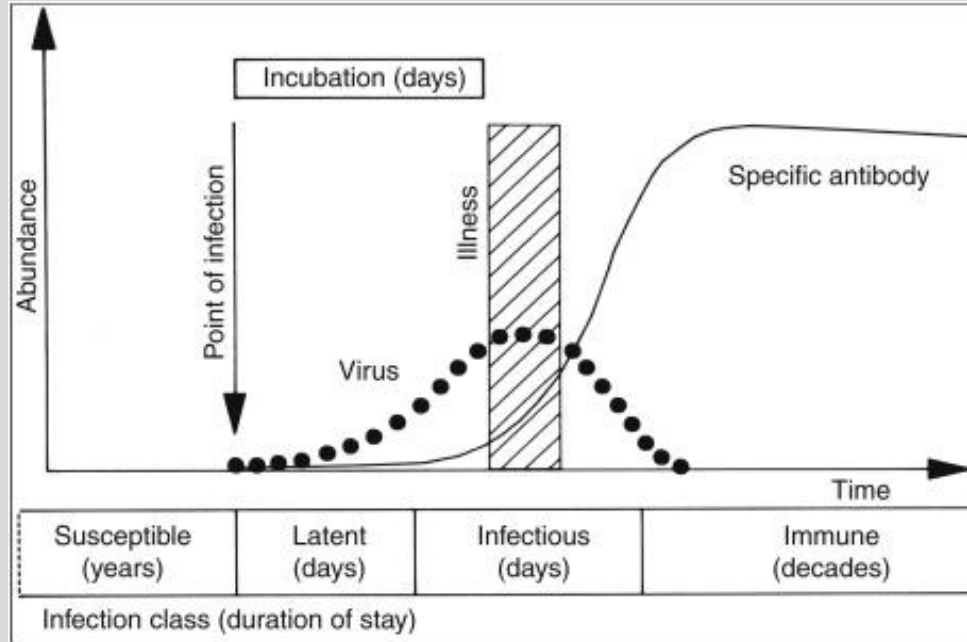


Why Vaccines Matter

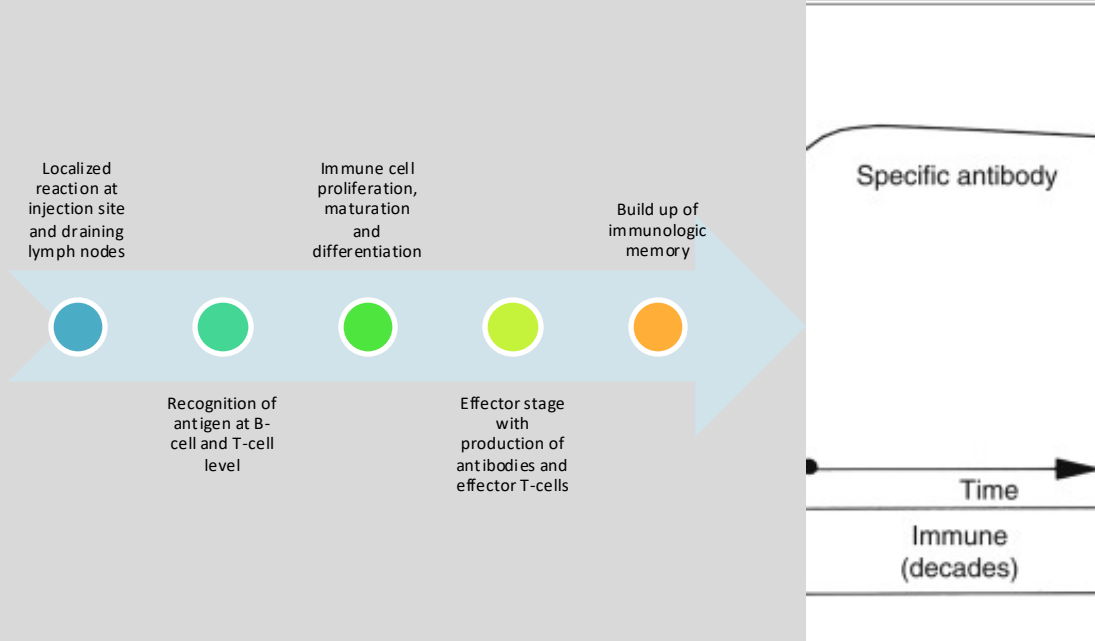
- Among the most effective interventions in medical history – prevents severe disease and transmission!
- Prevent morbidity, mortality, and disability resulting in the promotion of wellness throughout the lifespan
- Dramatic reduction healthcare costs
- Protect vulnerable populations who are unable to produce an immune response through herd effects
 - e.g., immune suppressed, contraindication to vaccines, refugees/overcrowding, low-income countries
- Prevention or attenuation of outbreaks, epidemics, pandemics

Typical Course of Viral Infections



- Immunity occurs either via infection or immunization
- Primary infection without immunity results in disease and sequelae
 - Slow onset of primary response
 - Lower antibody titer
 - Initial IgM -> IgG class switch
- Transmission occurs until primary immunity controls infection

Vaccines Alter the Course of Infection



- Immunization primes the immune system to reduce or prevent disease, sequelae and transmission
- Secondary response
 - Rapid, more robust memory response
 - Higher affinity antibodies
 - Higher IgG titers

Vaccine-preventable Complications

- Diphtheria: upper airway obstruction, systemic toxin-mediated illness, myocarditis, arrhythmia, heart failure, peripheral neuropathy, renal injury HiB: otitis media, epiglottitis, **meningitis**, bacteremia, sepsis, septic arthritis, pneumonia, **hearing loss**, **cognitive impairment**/development delay, **seizures**
- HepA: fulminant hepatic failure (rare), cholestatic hepatitis
- HepB: chronic hepatitis, **liver failure**, **cirrhosis**, **hepatocellular carcinoma**, HepB-associated glomerulonephritis and polyarteritis nodosa
- HPV: recurrent respiratory papillomatosis, genital warts, **cancers of cervix, anus, penis, oropharynx, vulva/vagina**
- Influenza: viral pneumonia, secondary bacterial pneumonia, otitis media (children), **myocarditis**, encephalopathy, **asthma exacerbation**, **COPD exacerbation**, **myocardial infarction**, **stroke**
- Measles: viral pneumonia, secondary bacterial pneumonia, acute measles encephalitis, SSPE, post-measles immune suppression associated secondary infection
- Meningococcus: meningitis, septic shock, purpura fulminans and limb loss, hearing loss, permanent neurologic impairment, chronic kidney injury
- Mumps: aseptic meningitis, orchitis, infertility, oophoritis, mastitis, sensorineural hearing loss
- Pertussis: lower respiratory infection, apnea and hypoxic brain injury (infants), pneumonia, seizures, encephalopathy
- Pneumococcal: invasive pneumonia, **sepsis**, **septic shock**, **meningitis**, otitis media (children), hearing loss
- Polio: acute poliomyelitis, irreversible flaccid paralysis, respiratory failure due to respiratory muscle paralysis, **post-polio syndrome**
- **Rotavirus**: dehydration, shock, electrolyte derangements
- RSV: hypoxic respiratory failure, asthma exacerbation, **reductions in all-cause mortality in older adults**
- Rubella: **congenital rubella syndrome** (cataracts, deafness, patent ductus arteriosus, pulmonary artery stenosis, microcephaly, developmental delay)
- SARS-CoV-2: carditis, long-COVID
- Tetanus: toxin-mediated spastic paralysis, laryngospasm and respiratory failure, autonomic instability, aspiration pneumonia, fractures, rhabdomyolysis
- Varicella: viral pneumonia, bacterial skin superinfection, necrotizing fasciitis, encephalitis, cerebellitis, **herpes zoster**, **postherpetic neuralgia**, vision loss (herpes zoster ophthalmicus)

Vaccine-preventable Complications

- **MMR** → prevents encephalitis + SSPE + congenital rubella syndrome
- **DTaP/Tdap** → prevents airway obstruction (diphtheria), spasms (tetanus), hypoxic brain injury (pertussis)
- **Hib/PCV/MenACWY/MenB** → prevents meningitis → prevents hearing loss + neurologic disability
- **IPV** → prevents irreversible paralysis, post-polio syndrome
- **HepB** → prevents chronic hepatitis → cirrhosis → HCC
- **HPV** → prevents persistent infection → cervical/oropharyngeal cancer
- **Rotavirus** → prevents severe dehydration hospitalization/death
- **Influenza/COVID** → prevents severe pneumonia, death, and reduces cardiovascular/inflammatory sequelae
- **Shingrix** → prevents shingles → postherpetic neuralgia

Platforms Conferring Immunity

Vaccines (active)

- Live attenuated
- Inactivated (killed)
- Subunit/recombinant protein
- Toxoid
- Polysaccharide
- Conjugate
- mRNA
- Viral vector

Passive immunity

- **Instant effect**, poor durability, no effect on adaptive immunity
- Immunoglobulin preparations
 - Maternal IgG transfer
 - IVIG
 - Rabies immune globulin
 - HBIG
- Monoclonal antibodies
 - Palivizumab (RSV)
 - Casirivimab/imdevimab (SARS-CoV-2)

Live Attenuated & Inactivated Vaccines

Live attenuated

Examples: MMR, varicella, rotavirus, LAIV, YF, OPV, oral typhoid

- Replication of intact, weakened pathogen in host
- Can be cold-adapted (LAIV)
- **Strong, long-lasting** humoral and cellular immunity
- **Contraindicated** in pregnancy and immune compromised
 - (e.g., low CD4, transplant, cytotoxic chemotherapy)

Inactivated

Examples: IPV, hepA, rabies, inactivated influenza, whole killed pertussis

- Heat, radiation, chemical inactivation
- No intact pathogen = no replication
- Strong humoral and T-cell immunity
- Long-lasting immunity requires boosters/adjuvants
- **Can be given in pregnancy and to immune compromised**

Subunit & Toxoid Vaccines

Subunit

Examples: HepB, Shingrix,
Novavax COVID

- Only specific antigens included
- Proteins or glycoproteins
- Manufactured from genetic code in producer cell
- Require adjuvants/boosters
- Excellent safety profile

Toxoid

Examples: tetanus, diphtheria

- Inactivated bacterial toxin
- Prevents toxin-mediated disease
- Primarily humoral immunity to toxin
- Bacterial growth unaffected
- Requires boosters

mRNA & Viral Vector Vaccines

mRNA

Examples: Moderna/Pfizer COVID-19,
Moderna CMV

- Lipid nanoparticle carrier delivers mRNA segment to local lymph nodes
- Antigen made rapidly in host cell cytoplasm
- Strong, durable antibody + T-cell response
- Low manufacturing cost
- **Rare carditis**

Viral Vector

Examples: J&J COVID, Ebola

- Modified, harmless virus delivers antigen gene
- **Strong cellular immunity**
- Pre-existing immunity to vector may reduce effectiveness
- Rare thrombosis with thrombocytopenia syndrome

Vaccine Contraindications

- Severe allergic reaction (eg, anaphylaxis) to a component of the vaccine
- Known severe immunodeficiency for patients and their close contacts (live attenuated vaccines)
- History of intussusception (rotavirus vaccine)
- Pregnancy (varicella vaccine, live-attenuated influenza vaccine, MMR)

Side Effects of Common Vaccines

Vaccine	Common Side Effects (Mild)	Rare or Serious Adverse Events
Influenza (Flu)	Soreness/redness at injection site, low-grade fever, muscle aches, headache.	Guillain-Barré Syndrome (GBS) (approx. 1-2 cases per million); Anaphylaxis.
MMR (Measles, Mumps, Rubella)	Fever, mild rash, swollen glands in cheeks or neck.	Febrile seizures (approx. 1 in 3,000); Temporary low platelet count.
DTaP / Tdap (Tetanus, Diphtheria, Pertussis)	Pain/swelling at site, fussiness, tiredness, loss of appetite.	Permanent brain damage (extremely rare, link is debated); Severe swelling of the entire limb.
HPV (Gardasil 9)	Pain/redness at site, headache, nausea, dizziness.	Syncope (fainting)—especially in adolescents; clinical protocol suggests 15-min observation.
Varicella (Chickenpox)	Sore arm, mild rash (up to 1 month after), joint stiffness.	Pneumonia; Seizures (very rare).
Hepatitis B	Soreness at injection site, temperature of 99.9° F or higher.	Severe allergic reaction (anaphylaxis) to yeast (used in production).
Shingles (Shingrix)	Shivering, muscle pain, fatigue, upset stomach.	Guillain-Barré Syndrome (GBS) (rare risk in seniors).
Pneumococcal (PCV15/20)	Swelling/redness at site, loss of appetite, irritability.	High fever; Allergic reactions to vaccine components.

Efficacy versus Effectiveness

- **Efficacy:** reduction of disease in controlled trials
- **Effectiveness:** real-world performance (population-based)
- **Why They Differ**
 - adherence to dosing schedule
 - storage/handling errors
 - host immune status
 - pathogen evolution
 - socioeconomic barriers
- **Outcomes Measured**
 - infection prevention
 - reduction in severe disease
 - prevention of complications
 - reduction in transmission

Influenza Vaccine

- Inactivated or recombinant (typically)
- Neutralizing antibodies against hemagglutinin (HA) prevent cell entry preventing reduce infection severity
- Protection varies due to antigenic drift; is absent after antigenic shift
- Expect 50-70% effectiveness

Figure 6. Flu Vaccination Coverage by Racial/Ethnic Group, Adults 18 years and older, United States, 2010–2021

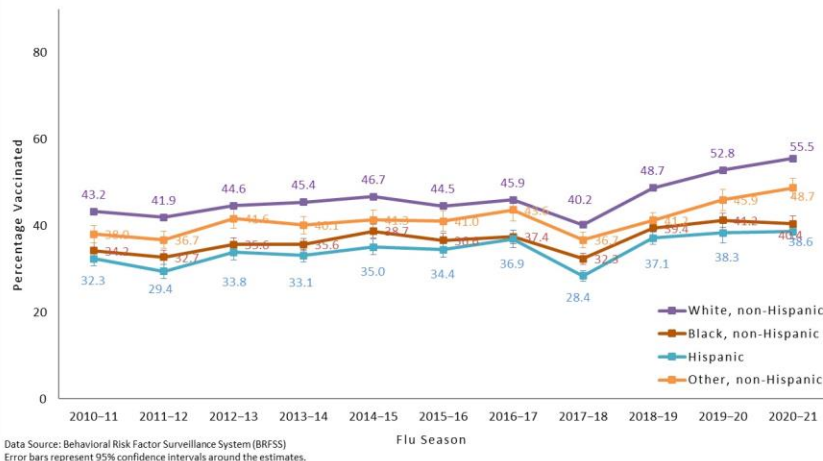
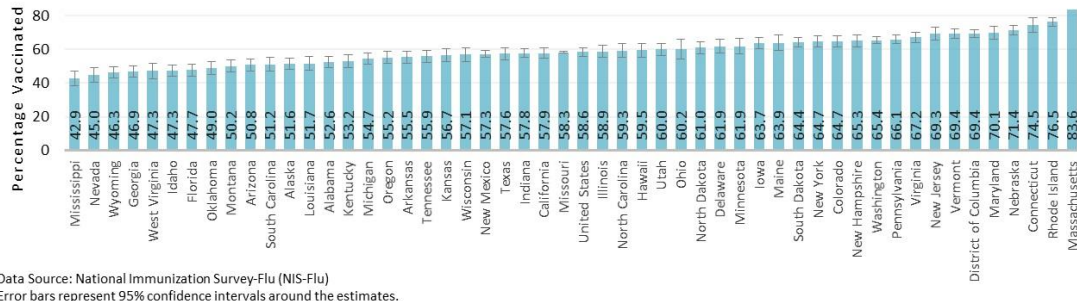


Figure 2. Flu Vaccination Coverage by State, Children 6 months–17 years, United States, 2020–21 Season



Measles Vaccine

- Live attenuated virus that induced **neutralizing antibodies** + memory T cells
- Highly effective after 2 doses
- High herd immunity threshold (95%)
- Prevented 23 million deaths between 2010 and 2018 (WHO)
 - Measles pneumonia, SSPE, post-measles immune suppression

Vaccine Truths

1. Vaccines do not prevent infection, they reduce the risk of severe disease, death, sequelae and complications
2. Inactivated or component vaccines do not replicate or cause disease
3. Some vaccines result in immune protection that is more robust than infection
4. Current vaccines were either studied in placebo-controlled trials or were studied in comparison to vaccines that were
5. With exception of SARS-CoV-2 (the pandemic drove economic development), vaccines for infections do not provide significant revenue for pharma
6. Side effects of vaccines, in general, are less prevalent (sometimes dramatically) than similar effects of infection
7. The effectiveness of unstudied, alternate schedules cannot be predicted by studied schedules
8. Eradication of disease does not eradicate circulating virus in all cases (e.g., polio)
9. Antigen exposure from vaccine is extremely limited (1-5) compared to the thousands of exposures of daily life
10. Vaccines do not enter the host cell nucleus or alter DNA; infectious agents do
11. No currently U.S. licensed vaccine contains aluminum, nor has aluminum in vaccines been linked to any disease process
12. No currently U.S. licensed vaccine contains thimerosal, nor has thimerosal in vaccines been linked to any disease process
13. Vaccines do not promote promiscuity or dangerous behavior
14. Anticancer vaccines are now being introduced for treatment